Logic, parity and simple probability

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L1-CO2 Level 1 Foundations · Combinatorics

Solve truth-teller and liar puzzles, use odd and even numbers and the pigeonhole principle to prove things impossible or certain, and find probabilities by counting.
By the end you can solve truth-teller and liar puzzles by testing cases and checking every statement; use parity and the pigeonhole principle to show that something is impossible or certain; find simple probabilities by counting equally likely outcomes.
Prerequisites: Listing and counting systematically (L1-CO1). **Contests:** JMC, Junior Kangaroo, JMO, AMC 8. **Time:** about 60 minutes.

1 Lesson

Ravi, Sara and Tom each either always tell the truth or always lie. Ravi says, "Sara is lying." Sara says, "Tom is lying." Tom says, "Ravi and Sara are both lying." Who is telling the truth?

We make an assumption and follow the assumption. Suppose Ravi tells the truth. Then Sara is lying, so Tom is not lying, so Tom's statement is true and Ravi is lying. This contradicts the assumption, so Ravi is a liar. Then Ravi's statement is false, so Sara tells the truth. Hence Tom is lying, and his statement is false, since Sara is not lying. Every statement agrees with its speaker's type, so only Sara tells the truth.

1.1 Truth-tellers and liars

Method.

1. Assume the type of one person, usually one whose statement mentions several others.
2. Follow the consequences until every person's type is decided.
3. Check every statement. If one contradicts the assumption, the assumption was false, so try the other type for that person.
4. If both cases survive, the question can only be asking about something they have in common.

Two facts shorten the work. If A says "B is a liar", then A and B are of different types. If A tells the truth, then B is a liar, and if A lies then B tells the truth. If A says "B tells the truth", then A and B are of the same type.

1.2 Parity

Every whole number is either even or odd. This is called its *parity*. For addition,

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$$\text{even} + \text{even} = \text{even}, \qquad \text{odd} + \text{odd} = \text{even}, \qquad \text{even} + \text{odd} = \text{odd}.$$

These hold because an odd number is an even number plus 1, and two spare 1s make an even number.

> A sum of whole numbers is odd exactly when an odd number of them are odd. A product of whole numbers is odd exactly when every one of them is odd.

Subtraction behaves like addition, since $a - b$ and $a + b$ differ by $2b$, which is even. Parity is most often used to show that something is impossible. If a quantity starts odd and every allowed move changes the quantity by an even number, the quantity stays odd for ever, and if every move changes the quantity by an odd number, its parity switches at every move.

1.3 The pigeonhole principle

> **Pigeonhole principle.**
> If more than n objects are put into n boxes, then some box contains at least two objects. Similarly, if more than $2n$ objects are put into n boxes, some box contains at least three, and so on.

The proof is one line. If every box held at most one object, there would be at most n objects. So among any 13 people, two were born in the same month (the 12 months are the boxes), and among any 25 people, three were.

Contest questions usually ask for the smallest number of objects that makes something *certain*. Imagine the unluckiest possible choices, find the largest number you could take without success, and add 1.

1.4 Probability

> If an experiment has a number of *equally likely* outcomes, the probability of an event is
>
> $$\frac{\text{number of outcomes in which the event happens}}{\text{total number of outcomes}}.$$

A probability lies between 0 (impossible) and 1 (certain), and the probability that an event does not happen is 1 minus the probability that the event does. The hard part is choosing outcomes that really are equally likely. For two dice, the 11 totals from 2 to 12 are not, but the 36 ordered pairs (first score, second score) are.

> **Example 1.**
> Five people each either always tell the truth or always lie. Amy says, "At least two of us are liars." Bilal says, "At most three of us are liars." Chen says, "The number of liars among us is even." Dina says, "Exactly one of us tells the truth." Eli says, "There are more truth-tellers than liars among us." Who are the liars?
>
> *Solution.*
> Every statement is about the whole group, so assuming one person's type settles little. Instead, let k be the number of liars, which is one of 0 to 5. Once k is known we can say which

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statements are true, and since the liars are exactly the people whose statements are false, the number of false statements must be k.
In terms of k, Amy's statement is true when k is at least 2, Bilal's when k is at most 3, Chen's when k is 0, 2 or 4 (0 is even), Dina's when $k = 4$ (one truth-teller means four liars), and Eli's when k is 0, 1 or 2. So the false statements are Amy's and Dina's when $k = 0$ (2 of them), Amy's, Chen's and Dina's when $k = 1$ (3), only Dina's when $k = 2$ (1), Chen's, Dina's and Eli's when $k = 3$ (3), Bilal's and Eli's when $k = 4$ (2), and all but Amy's when $k = 5$ (4). Only $k = 3$ gives k false statements.
Check that with three liars, Amy and Bilal speak the truth, while Chen, Dina and Eli do not, since 3 is odd and there are two truth-tellers and three liars. So the liars are Chen, Dina and Eli.

Example 2.
A robot starts at 0 on a number line. Every minute the robot moves either 3 to the right or 5 to the left. Can the robot be back at 0 after exactly 15 minutes? After exactly 8 minutes?

Solution.
Experiment first. After one minute the robot is at 3 or -5, and after two minutes at 6, -2 or -10. The positions after one minute are odd and those after two are even, which suggests looking at parity. (Negative numbers are odd or even in the same way, as $-5 = -6 + 1$ is odd.)
Each move adds 3 or takes away 5, and both are odd, so every move changes the parity of the robot's position. The robot starts at 0, which is even, so its position is odd after 1, 3, 5, ... minutes and even after 2, 4, 6, ... minutes. After 15 minutes the robot is at an odd number, so the robot cannot be at 0.
For 8 minutes, parity does not rule out 0, but that alone does not show that 0 can be reached. We need an example. Five moves right and three moves left take the robot $5 \times 3 = 15$ to the right and $3 \times 5 = 15$ to the left, so the robot ends at 0. The answers are no and yes. Parity can show that something is impossible. To show that something is possible, give an example.

Example 3.
A bag contains 5 red, 4 blue and 3 green marbles. Marbles are taken out without looking. How many must be taken to be certain of having (a) two marbles of the same colour, (b) three marbles of the same colour, (c) at least one marble of each colour?

Solution.
In each part we ask how many marbles could be taken while still failing, in the unluckiest case.
(a) Three marbles could be one of each colour, so 3 is not enough. With 4 marbles and 3 colours, two marbles have the same colour by the pigeonhole principle, with the colours as boxes. The answer is 4.
(b) Six marbles could be two of each colour, so 6 is not enough. Among 7 marbles some colour appears three times, since otherwise there would be at most $2 + 2 + 2 = 6$ marbles. The answer is 7.
(c) Here the unluckiest case is different. You could take all 5 red and all 4 blue marbles, 9 in total, and still have no green. After that only green marbles are left, so the tenth marble is green and the answer is 10. Copying the reasoning of (a) here is the usual mistake.

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Example 4.
Two fair six-sided dice are thrown. What is the probability that the two scores differ by exactly 2?

Solution.
The pairs of scores that differ by 2 are 1 and 3, 2 and 4, 3 and 5, and 4 and 6. Many people compare these 4 pairs with the 21 possible pairs of scores (6 doubles and 15 others) and answer $\frac{4}{21}$. But a 1 and a 3 can happen in two ways (1 on the first die and 3 on the second, or the other way round), while a double 1 happens in only one way, so the 21 pairs are not equally likely. Instead we record each outcome as (first score, second score). There are $6 \times 6 = 36$, all equally likely.

first die \ second die	1	2	3	4	5	6
1	1,1	1,2	1,3	1,4	1,5	1,6
2	2,1	2,2	2,3	2,4	2,5	2,6
3	3,1	3,2	3,3	3,4	3,5	3,6
4	4,1	4,2	4,3	4,4	4,5	4,6
5	5,1	5,2	5,3	5,4	5,5	5,6
6	6,1	6,2	6,3	6,4	6,5	6,6

The shaded outcomes are those with scores differing by 2. There are 8 of them, each of the four pairs above in two orders, so the probability is $\frac{8}{36} = \frac{2}{9}$.

Common traps.

- In a liar puzzle, finding a contradiction in one case and not checking that the other case works.
- Thinking that "A says B is a liar" shows that A tells the truth. The statement shows only that A and B are of different types.
- In "to be certain" questions, answering with the luckiest case instead of the unluckiest, or forgetting to add 1.
- Counting outcomes that are not equally likely, such as the totals of two dice.
- Forgetting that 0 is even.

1.5 How to spot this in a contest

1. "Always tells the truth" or "always lies". Assume one person's type and follow the consequences. If every statement is about how many liars there are, try each possible number of liars instead.
2. "Is ... possible?" after a repeated move. Find a quantity whose parity one move never changes, or always changes. For moves between neighbouring squares, colour the grid like a chessboard.

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3. "How many must be taken to be sure". Use pigeonhole, after imagining the unluckiest case.
4. "What is the probability". Count equally likely outcomes, with a 6 by 6 table for two dice.
5. Sums or products and the words "odd" or "even". Count how many of the numbers are odd.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ The eleven letters of the word PROBABILITY are written on eleven tiles, one letter on each tile, and the tiles are put in a bag. One tile is taken out at random. What is the probability that the tile shows a B or an I?

Problem 2 ⋆ What is the smallest number of pupils a class must have to be certain that at least three of them were born on the same day of the week?

Problem 3 ⋆⋆ Asha, Ben and Cara each either always tell the truth or always lie. Asha says, "Ben and Cara are both liars." Ben says, "Cara is a liar." Cara says, "Exactly two of us three tell the truth." Who tells the truth?

Problem 4 ⋆⋆ Two fair six-sided dice are thrown. What is the probability that the larger of the two scores is 4? (If both dice show 4, the larger score counts as 4.)

Problem 5 ⋆⋆⋆ In a year that is not a leap year, the months have 31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30 and 31 days. A plus sign or a minus sign is placed in front of each of these twelve numbers, and the resulting twelve numbers are added. Explain why the total can never be 0.

Problem 6 ⋆⋆⋆ A classroom has 25 desks arranged in a 5 by 5 grid. A cleaner wants to wipe every desk exactly once. She starts at a corner desk, and after each desk she moves to a desk directly in front of, behind, to the left of or to the right of her current desk. Explain why the last desk she wipes cannot be next to the corner desk where she started.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **UKMT JMC 2015 Question 17** ⋆
   Each Knave after the first speaks about the one before, so are the two the same type or different types?
2. **UKMT Junior Kangaroo 2019 Question 11** ⋆
   If one of the facts is true, which others must also be true?
3. **AMC 8 2022 Problem 12** ⋆
   16 equally likely outcomes. List the squares in the possible range
4. **AMC 8 2017 Problem 10** ⋆⋆
   Count the sets of three cards, then the sets whose largest card is 4

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5. **AMC 8 2017 Problem 8** ⋆⋆
   Look for two statements that cannot both be true of a two-digit number

6. **UKMT JMC 2018 Question 20** ⋆⋆
   Imagine the unluckiest sequence of picks, then add one

7. **AMC 8 2019 Problem 18** ⋆⋆
   When is a sum of two numbers even? Count the odd and even faces

8. **UKMT JMC 2022 Question 20** ⋆⋆
   Start with Pam and Quentin. Are they the same type or different types? Then test Susan and Terry

9. **UKMT Junior Kangaroo 2024 Question 16** ⋆⋆⋆
   For the unluckiest case, how much fruit can be taken with no apple and pear of the same colour?

10. **UKMT Junior Kangaroo 2018 Question 14** ⋆⋆⋆
    When is a sum of two numbers even? What must the four cards Avani has not taken look like?

11. **UKMT JMC 2024 Question 19** ⋆⋆⋆
    Assume P is a Clown and follow the statements. Then assume P is a Joker

12. **UKMT JMC 2026 Question 11** ⋆⋆⋆
    Add up the three row sums. What is the parity of the result?

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4 Hints

1. Each tile is equally likely to be taken. How many tiles show a B or an I?
2. What is the largest class in which no day of the week is the birthday of three pupils?
3. Assume first that Cara tells the truth. Ben's statement tells you how Ben and Cara are related.
4. List the outcomes as ordered pairs (first score, second score). Be careful not to count the double 4 twice.
5. Start with every sign a plus. What happens to the total, and to its parity, when one sign is changed?
6. Colour the desks like a chessboard. What happens to the colour each time she moves?

5 Solutions

Problem 1

The eleven letters of the word PROBABILITY are written on eleven tiles, one letter on each tile, and the tiles are put in a bag. One tile is taken out at random. What is the probability that the tile shows a B or an I?

There are 11 tiles, each equally likely to be taken. The word PROBABILITY contains two Bs and two Is, so 4 tiles show a B or an I. The probability is $\frac{4}{11}$.

Answer: $\frac{4}{11}$

Problem 2

What is the smallest number of pupils a class must have to be certain that at least three of them were born on the same day of the week?

The seven days of the week are the boxes. A class of 14 could have exactly two pupils born on each day, so 14 pupils are not enough. In a class of 15, if no day had three or more pupils, each day would have at most two, which makes at most $7 \times 2 = 14$ pupils. This is a contradiction, so some day has at least three. The answer is 15.

Answer: 15

Problem 3

Asha, Ben and Cara each either always tell the truth or always lie. Asha says, "Ben and Cara are both liars." Ben says, "Cara is a liar." Cara says, "Exactly two of us three tell the truth." Who tells the truth?

Ben says that Cara is a liar, so Ben and Cara are of different types.

Suppose Cara tells the truth. Then exactly two of the three tell the truth, and Ben does not, so Asha does. But Asha says that Cara is a liar, which is false. This is a contradiction, so Cara is a liar.

Then Ben tells the truth. Asha's statement says that Ben is a liar, which is false, so Asha is a liar too. Finally we check Cara. Only Ben tells the truth, so Cara's statement is false, as required for a liar. So only Ben tells the truth.

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Answer: Only Ben

Problem 4

Two fair six-sided dice are thrown. What is the probability that the larger of the two scores is 4? (If both dice show 4, the larger score counts as 4.)

Record each outcome as (first score, second score). There are $6 \times 6 = 36$ equally likely outcomes. The larger score is 4 when one die shows 4 and the other shows at most 4. With the first die showing 4 there are 4 outcomes, (4,1), (4,2), (4,3) and (4,4). With the second die showing 4 and the first less than 4 there are 3 more, (1,4), (2,4) and (3,4). The outcome (4,4) has been counted only once, so there are 7 favourable outcomes, and the probability is $\frac{7}{36}$.

A second way is to note that both scores are at most 4 in $4 \times 4 = 16$ outcomes, and both are at most 3 in $3 \times 3 = 9$ of those. The larger score is 4 in the other $16 - 9 = 7$.

Answer: $\frac{7}{36}$

Problem 5

In a year that is not a leap year, the months have 31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30 and 31 days. A plus sign or a minus sign is placed in front of each of these twelve numbers, and the resulting twelve numbers are added. Explain why the total can never be 0.

If every sign is a plus, the total is the number of days in the year, 365, which is odd. (This can be seen without adding, since seven of the twelve numbers are odd and a sum is odd when an odd number of its terms are odd.)

Changing the sign in front of a number n from plus to minus lowers the total by $2n$, since n is taken away instead of added. As $2n$ is even, this does not change the parity of the total. Any choice of signs can be reached from all plus signs by changing signs one at a time, so the total is always odd.

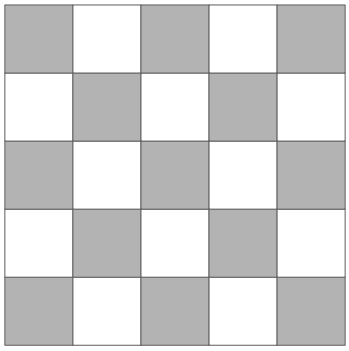
Since 0 is even, the total can never be 0.

Problem 6

A classroom has 25 desks arranged in a 5 by 5 grid. A cleaner wants to wipe every desk exactly once. She starts at a corner desk, and after each desk she moves to a desk directly in front of, behind, to the left of or to the right of her current desk. Explain why the last desk she wipes cannot be next to the corner desk where she started.

Colour the desks like a chessboard, with the corner desks shaded, as in the diagram. Two desks next to each other in a row or column always have different colours, so the colours of the desks she wipes alternate, going shaded, unshaded, shaded, and so on.

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She starts at a shaded corner desk, so the 1st, 3rd, 5th, ... desks she wipes are shaded and the 2nd, 4th, 6th, ... are unshaded. She wipes 25 desks, and 25 is odd, so the last desk is shaded. A desk next to the starting corner has the other colour, so that desk cannot be the last one.

As a check, the shaded desks she wipes are the 1st, 3rd, ..., 25th, which is 13 desks, and the grid does have 13 shaded desks (the rows contain 3, 2, 3, 2 and 3), with 12 unshaded.